

DTW based Authentication for Wireless Medical Device Security

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Abstract—Wireless medical devices play an important role in providing safety and privacy to patients suffering from major health issues. These light-weight devices can be worn inside or outside the patient's body and provide more convenience and reliable doctor-patient communication. However, the design, development, and usage of these devices play a critical role in present network paradigm. They are vulnerable to network threats and attacks which break the confidentiality, integrity and availability protocols in networking scenarios. Thus, it is important to have identification and authentication of only the authorized people who can operate the device. This paper proposes Dynamic Time Warping (DTW) algorithm for providing trusted authentication and identification of only authorized people using ECG signal. Here, DTW algorithm is used to measure the correlation between different ECG signal records. Experiments were carried out to evaluate the proposed algorithm with a large database consisting of users of all ages, including abnormal ECG data and long span of time intervals between ECG recordings for evaluating the reliability of the proposed algorithm. Comparative evaluation of the proposed system showed that, it is not only efficient, but also light weight in comparison to the existing systems.

Keywords: Security, DTW, dynamic time warping, machine learning, insulin pumps, deep learning, implantable medical devices.

I. INTRODUCTION

Wireless monitoring of medical devices has given many hospitals the ability to provide continuous patient care throughout the treatment process. Given its potential, usage of such devices is growing at a fast pace. It is reported that there are more than 6500 medical device companies in the United States [1]. The global wearable medical device (WMD) market size was valued at USD 3.9 billion in 2014 and is expected to grow at the rate of 28.0% over the next few years [2].

Given the rapid adoption of WMDs, it is critical to ensure that they are connected securely when dealing with patient's sensitive data. Security in these devices plays a major role since it directly affects the patient's health and personal information. Given this requirement, security in these devices should address three major goals viz confidentiality, integrity and availability [3].

- *Confidentiality* ensures that the devices are accessed only by authorized people. At the same time, it plays a major role in secrecy and privacy. Confidentiality breaches can

introduce attacks such as eavesdropping and impersonating other individuals.

- *Integrity* is an important goal which means that devices should be modified only by authorized people and in approved ways. If this feature is not present, an adversary in this case, can introduce the man-in-the-middle attack and message replay attack to modify the settings of the device which can exhaust the resources of the device.
- *Availability* means that the devices are accessible to only authorized people at appropriate times. Denial of service attack is one major attack, which can exhaust the device network bandwidth and computing resources in order to congest the network and make device non-functional.

Authentication enforces a security policy that specifies which individuals can access resources and objectives of the medical devices. This policy ensures and assures the accuracy of identities.

In the literature, several authentication security solutions have been proposed for WMDs ([5]- [17]). These solutions mainly utilize biometric approaches, distance/proximity based techniques, key management protocols and external device methodologies [13]. However, where biometric-based approaches are unable to accommodate changes with respect to time; distance based techniques and external device methodologies provide weak authentication since the adversary can come close to the patient to introduce an attack. This has led to an urgent requirement to introduce an efficient and accurate authorization technique for WMDs.

This paper presents an efficient and reliable authentication mechanism using Dynamic Time Warping (DTW) technique through electrocardiogram (ECG) signals to give permission to authorised people. DTW is an effective classifier that works extremely well to recognize temporal motions. Temporal motions such as ECG can be characterized as a strong arrangement of developments that happen over a varying time period. ECG signals lend themselves as ideal candidate for DTW technique due to their unique wave shape, which is derived from the anatomical structure of the heart and physiological conditions. Due to this, we utilize ECG signal for DTW to provide authentication to authorized people for WMD security. The paper outline is as follows; Section II presents the current state of the art in the domain of WMD security.

Section III proposes a DTW based authentication mechanism to be applied to WMD security. Section IV presents the results of the proposed model and compares these results with the existing solutions. Section V presents conclusions and scope for future work.

II. RELATED WORK

Security in WMD plays a crucial role since it is directly related to patients' health. Recently there have been numerous cases wherein, attackers threaten the victims by attacking the device in order to acquire ransom [4] from the patients. It is thus an urgent requirement to develop techniques which achieve a high level of security for medical devices. The main objective of installing a medical device inside the body is to enable doctors with regular and remote access of the patient. Subsequently, the doctor is able to access the information anytime and anywhere. Likewise, the patient should be able to access and perform operations on the device as and when it is required. One way to achieve this is by requiring authentication in WMD as it is aimed at validating a claimed identity of an end user or a device. Only approved, users ought to be permitted to adjust medical devices or to generally trigger particular device conduct (for instance, an attacker should not have the capacity to trigger pacemaker test mode, which could instigate heart collapse). Doctors or device makers ought to place limits on the settings accessible to patients to keep from inadvertently or deliberately hurting themselves (for example, patients ought not have the right to inject bolus injection into the embedded device). Additionally, while the doctor can have admittance to change most gadget settings, they should not have unlimited access to the review logs or investigate modes. It is also critical that WMDs acknowledge only approved firmware redesigns. Authorization of WMD is also important as it is an act of granting access rights to the user which can range from personal authorization to role-based medical device selection. Personal authorization of the device is given to specific set of authorized users who can use the device after manufacturing. This set includes patients, doctors or primary entities who can access the device for monitoring. Role-based authorization is prescribed to the device manufacturer who has role-based access to the device. When a third party, such as a wearable device or a mobile phone, has access to the WMD, it has the authorization to give them access. Thus, authorization and authentication in a medical setup can be sensitive to the setting on the device.

Security approaches in medical devices can be categorized into six types namely [13]: bio-metric based approaches, distance/proximity based techniques, key management protocols, anomaly detection mechanisms, audit mechanisms and external device methodologies. Biometric-based approaches take unique biological characteristics of individuals, such as retina scan [9], heart signals [10], to provide authentication. These approaches, which are based on changing attributes, do not fit for authentication purposes. Distance-based approaches estimate the distance between the WMD and caregiver by measuring the sent and received transmissions through techniques

such as proximity to piezoelectric element [18], Diffie-Helman protocol [21], and near field communication [22]. These approaches provide weak authentication since the adversary can come close to the patient to breach the security provided by distance based techniques. Key management protocols exchange keys in order to provide authentication between patient and caregiver. Symmetric keys [18], public keys [19] and physiological signals [20] are used in medical devices for the generation of keys. However, key generation protocols provide less reliability due to the extra waiting time for the generation and exchange of keys. Audit based mechanism maintains a track of all the transaction logs that happen in the WMD [28]. However, this processing is too intensive for resource optimised medical devices. Anomaly based mechanisms [7], based on the access pattern of the medical device functionality, pass the pattern to Support Vector Machine (SVM) [9] to classify the individuals. SVM is considered as the most efficient authentication mechanism for medical devices. However, the technique is highly complex for resource optimised medical devices. External device methodologies are also utilized to provide authentication such as MedMon [24], IMDGaurd [23], IMDShield, Cloaker [25]. External device methodologies not only require users to carry an extra gadget besides the WMD, but also consume battery life of the medical device. Additionally, the adversary can easily come close to the patient and disrupt the functionality of the device. To overcome the challenges mentioned above, this paper proposes a DTW based authentication mechanism.

The DTW algorithm is a supervised learning algorithm that can be utilized to classify any kind of N-dimensional, temporal signal. The DTW algorithm works by making a template time series for each motion that could be perceived, and then warping the real time biometric signals to each of the templates to locate the best match. We applied rejection thresholds that enable the DTW algorithm to automatically reject values that are not within the limit. Acceptance of ECG as a biometric signal is driven by the fact that it remains the best and the least intrusive technique for the authentication task it performs. ECG estimation frameworks are widely used since they more reliable, are able to perform a wider range of functions, and are less complex to use as time has advanced [30]. This paper uses ECG signals for providing authentication to authorized people based on a threshold devised by running the simulations over a large sample set.

III. PROPOSED WORK

A. Introduction: Dynamic Time Warping

Dynamic Time Warping (DTW) is an algorithm for aligning sequences with unique patterns varying over time intervals. It was originally used for speech recognition [31]. The algorithm produces best alignment path between two time sequences using a warping function based on time-normalized distance between the sequences. The objective of DTW is to compare and compute time warp distance between two time sequences. Let A and B be the reference and representative time series of length X and Y respectively where:

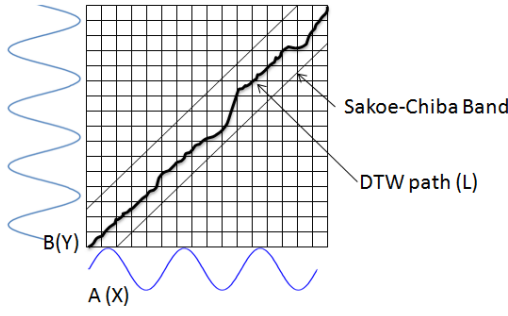


Fig. 1: Dynamic Time Warping: A and B are the two time series of length X and Y respectively. A DTW path is obtained of length L.

$$A = A_1, A_2, A_3, \dots, A_x, \dots, A_X, x \in N$$

$$B = B_1, B_2, B_3, \dots, B_y, \dots, B_Y, y \in N$$

In order to align the two time series, a x -by- y matrix is generated. The (x^{th}, y^{th}) element of the matrix correspond to the alignment distance which is a squared distance calculated using following equation [32]:

$$d(x, y) = (A_x - B_y)^2 \quad (1)$$

Overall distance between the two sequences is calculated using following equation where l is the distance of the warping path [33]:

$$D_i(A, B) = \sum_{l=1}^L d_l(A_{x_l}, B_{y_l}) \quad (2)$$

Using dynamic programming which is an iterative technique, an optimal warping path is found among all computed warping paths. The optimal warping path is the one that has the minimal overall distance or similarity score calculated using following equation [33]. This technique is shown in Figure 1:

$$DTW(A, B) = \min D_i(A, B) | \text{Warping path} \quad (3)$$

Furthermore, in order to reduce the number of warping paths for consideration, several constraints can be introduced during warping path computation:

- **Boundary Condition:** Path constraint is for starting and ending points where in the path starts from bottom left and ends at the top right.

$$x(1), y(1) = 1 \quad (4)$$

$$x(L) = X, x(L) = Y \quad (5)$$

where X, Y, L denote the length of the representative time series, sample time series and the warping path respectively.

- **Continuity Condition:** Path constraint is for the step count where in the path advances at most one step at a time.

$$x(l) - x(l-1) \leq 1 \quad (6)$$

$$y(l) - y(l-1) \leq 1 \quad (7)$$

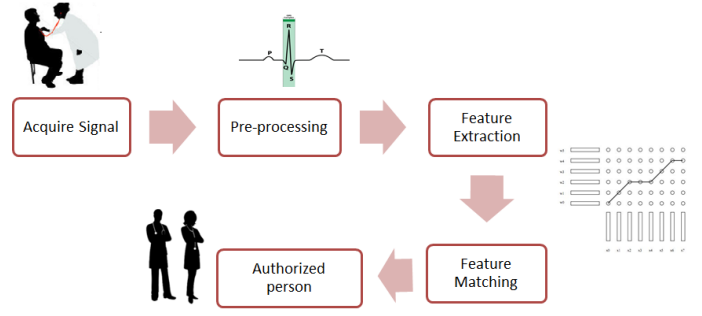


Fig. 2: DTW Authentication for Wireless Medical Device Security

- **Monotonic Condition:** Path either advances or stays the same but does not recede.

$$x(l) - x(l-1) \geq 0 \quad (8)$$

$$y(l) - y(l-1) \geq 0 \quad (9)$$

- **Slope Constraint Condition:** Path constraint is for the slope where in the path should not be too steep or too shallow.
- **Warping Window Condition:** Path is likely to be around the diagonal without deviating much from it.

$$|x(l) - y(l)| \leq w_l \quad (10)$$

where w_l is the window length. The Sakoe-Chiba band and Itakura Parallelogram [34] are two of the most prevalent warping window conditions being used.

This paper utilizes DTW mechanism to compute the minimum distance or the similarity score between the ECG signals to provide authentication to authorized people.

B. Description

In this paper, we propose ECG to be used as a biometric signal to uniquely identify a person. Initially, an ECG signal is acquired to perform pre-processing, followed by feature extraction and matching as shown in Figure 2. The pre-processing phase performs following steps before performing the feature extraction and matching:

- **Pre-emphasis:** The ECG signal is passed through a finite impulse response high pass filter which retains the magnitude of some higher frequencies with respect to the magnitude of other frequencies, thereby enhancing the over-all signal to noise ratio.
- **Divide signal into 20ms frame:** The input ECG signal is segmented into small frames of 20ms length with 50% overlap with the connecting frames to ensure coherence.
- **Windowing:** Each frame is multiplied with Hamming window [26] in time domain. This smoothens the discontinuity at the beginning and ending of every frame.
- **Noise threshold calculation:** For distinguishing any noise related to the signal acquisition from the input ECG signal, energy of each frame of the input signal is calculated and stored into an array. Size of the energy

array is equivalent to the total number of frames. This energy array is arranged in the ascending order and mean of initial 15 components is used as the energy of the noise. Signal energy is set to 10 times the noise energy.

- *True ECG signal detection*: Once the limit has been calculated, all the components in the energy array which are more prominent than the limit are marked as 1 and the rest as 0.

For the feature extraction phase, following steps are performed:

- *Fast Fourier Transform (FFT)*: FFT [29] is performed on each frame of the utterance.
- *Power Spectrum*: The most important aspect for ECG recognizer is the way in which the spectral shape of the ECG signal changes with time. Direct calculation of power spectrum from the ECG signal produces a spectrum containing “ripples” caused by the excitation spectrum. A smooth spectral shape that represents spectral shaping has to be assessed. This can be achieved by utilizing a low pass filtering technique. Inverse Fourier transform of the spectrum creates the time domain component, giving cepstrum.
- *Mel Frequency Warping*: Mel frequency warping is performed by utilizing triangle shaped filters in the spectral domain. This is done to calculate a weighted sum over the power spectrum coefficients which exist in every window. The spectrum of each frame is warped onto the Mel scale and consequently Mel spectral coefficients are obtained.
- *Discrete Cosine Transform*: Discrete cosine transform is done on Mel spectral coefficients of each frame, subsequently acquiring Mel Frequency Cepstral Coefficients (MFCC).

For the feature extraction phase, a dictionary containing feature vector sequence of the reference ECG signal is stored. The feature of the reference sequence is compared with each incoming signals in the sets using DTW. Using a pre-determined threshold, genuine ECG signals are differentiated from the false ECG signals. In the implementation of the program used in this paper, the threshold was set to 85.

IV. PERFORMANCE ANALYSIS

A. Dataset and Platform

To validate the proposed model, ECG-ID dataset from [35] is utilized. The database contains 310 ECG recordings, obtained from 90 people, each 20 seconds long, sampled at 500 Hz with 12-bit precision. The dataset contains two ECG signals: unfiltered (raw) and filtered. The raw ECG signal is noisy, therefore filtered ECG signal is used for DTW. The simulations were carried out using National Instruments LabVIEW 2015 and MathScript RT Module 2015.

B. Results

Extensive simulations were performed to determine a reliable zone in which we can set the threshold value to distinguish between authorised and unauthorised people. This would

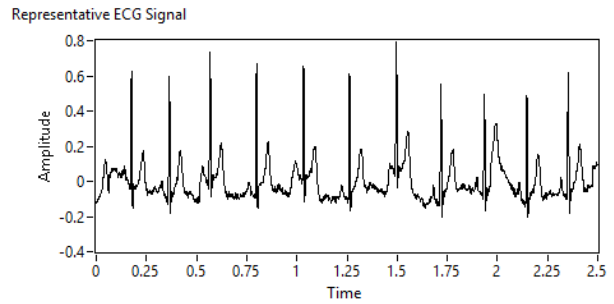


Fig. 3: A Representative Signal as Reference for DTW

then be used to grant authentication to authorized people. Figure 3 shows the ECG signal for Person 1 which is used as the reference biometric signal. Using the database provided by [35], a dataset containing 65 ECG waveforms was created. This dataset contains the waveforms as shown in Table I. As the test was executed, one waveform was randomly selected from the dataset.

TABLE I: Dataset Information

Record number	number of ECG waveforms
Person_01	15
Person_02	15
Person_03	5
Person_04	2
Person_05	2
Person_09	6
Person_15	2
Person_16	3
Person_24	5
Person_25	5
Person_30	5

The DTW algorithm was then used to calculate match confidence for each of these waveforms, with the ECG waveform shown in Figure 3 as the reference signal. Figure 4 shows the results of this simulation. The samples shown with green squares correspond to ECG waveforms from Person 1. The samples shown with red circles correspond to ECG waveforms from all other people, as described earlier. As can be seen from the results, there is a clear zone which distinguishes between Person 1 (authorized) from the rest of the people (unauthorized). Using this zone, one can establish a threshold value which separates correlated from uncorrelated signals.

The results shown in Figure 4 were based on experiments done with reference signals and ECG signals of length 20s. The effect of the length of the reference and ECG signals on the correlation parameter is studied next. To do so, we took three data sets. Similar to previous study, Record 0 of Person 0 was considered as the reference signal. Record 1 of Person 0 is considered for the correlated signal and Record 0 of Person 1 is considered for the uncorrelated signal. We got subsets of the 20s signals to create 6 different datasets of lengths 2s, 5s, 7s, 12s, 15s and finally 20s. In Figure 5(a), match confidence (%) is plotted for ECG signals of varying lengths (Time on x-axis) as a function of reference signals of different lengths. For

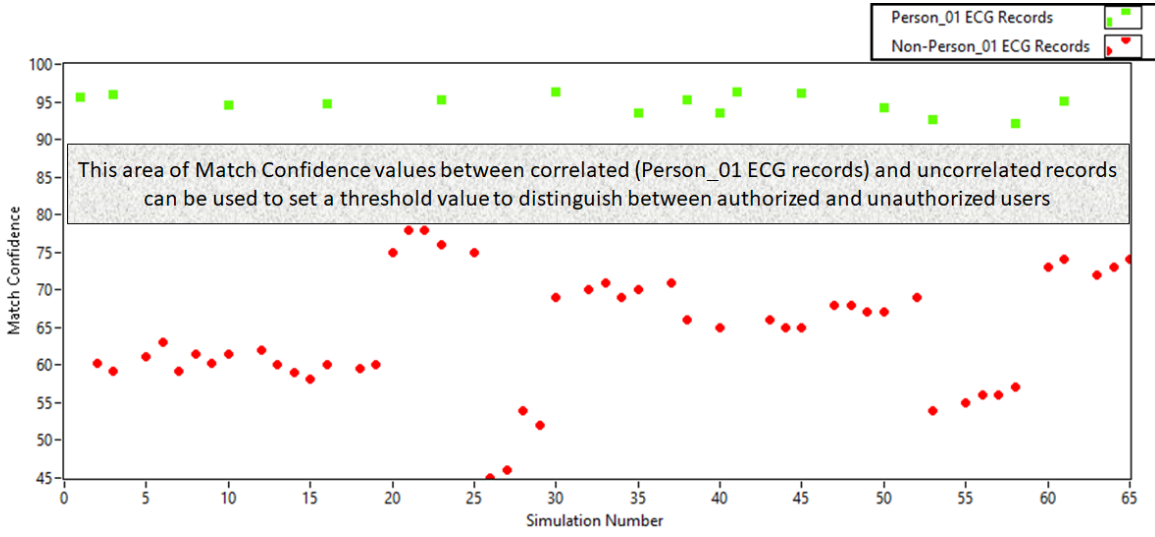


Fig. 4: Comparison of Authorized vs Unauthorized People: Here Patient 1 is taken as the reference signal and others as the test signal. Match confidence is more than 90% for patient 1 and less than 80% for other cases.

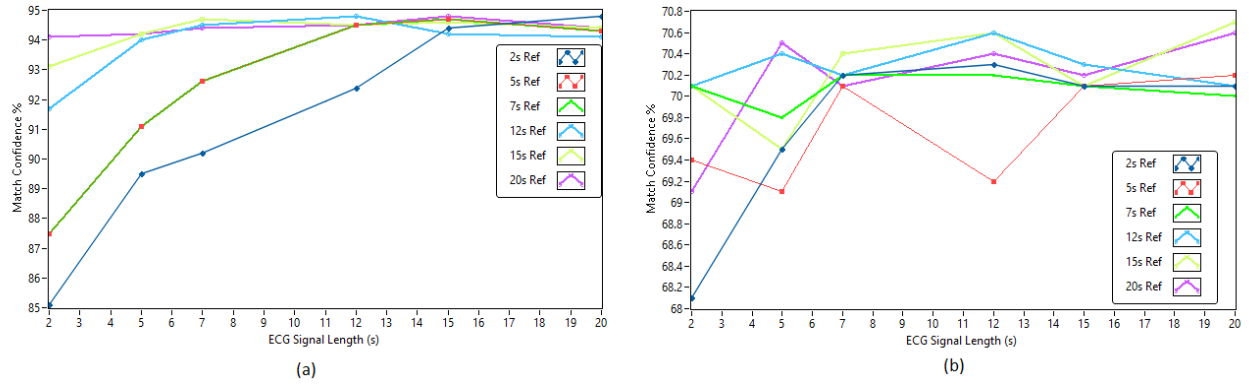


Fig. 5: Effect of Length on (a)Correlated Signals (b)Uncorrelated Signals

example, 2sRef is the graph of match confidence as a function of ECG signals length, when the reference signal is always 2 seconds long. Experiments shows that when the reference signal length is short, one requires a longer length ECG signal to achieve stable match confidence level. On the other hand, when the reference signal is longer (20sRef), match confidence does not vary across ECG signal lengths. Based on these findings, it is recommended to use a reference signal of longer length and ECG signals of shorter length. This operation is optimized for speed and memory utilization, as the reference signal is captured only once. Figure 5(b) shows similar results to compare with match confidence for uncorrelated signals. In this case, the effect of signal length is less pronounced as we vary the length of the reference signals.

C. Comparative Analysis

This section compares DTW authentication with other existing methods on two performance metrics, viz accuracy and complexity. Accuracy is defined as the ratio of correctly classified outcomes to the total number of outcomes. Com-

plexity is the measure of the behavior of the model which requires several interactions among the components to reach an optimal solution. Space and time complexity is defined as the amount of memory and time respectively required in the computation of the model. In this paper, we compare DTW with SVM [5]. DTW was nearly 99.9% accurate, since it was able to distinguish all the people while non-linear SVM achieved only 90.2% accuracy. Also, while space wise both were equally comparable, DTW was less complex time wise. As seen from the results, DTW showed better accuracy and complexity in comparison to SVM.

TABLE II: Comparative Analysis of DTW and SVM

Methodology	Accuracy	Space Complexity	Time Complexity
DTW	99.9%	$O(n^2)$	$O(n^2)$
Linear SVM	97.0%	$O(n^2)$	$O(n^3)$
Non-Linear SVM	90.2%	$O(n^2)$	$O(n^3)$

Note: n is the number of samples

V. CONCLUSIONS

This paper proposes an approach for individual authentication and validation utilizing ECG signals. Dynamic Time Warping (DTW) was used to classify genuine users from the unauthorized users. The proposal was tested on Physionet database size of 310 ECG recordings from 20 people. Our experimental results have presented a feasibility study of human identification and verification based on features extracted from ECG using DTW. The DTW technique was efficient in comparison to existing authentication mechanisms such as linear and non-linear SVM. As scope of future work, we plan to investigate the performance of the DTW algorithm on hardware platforms.

VI. ACKNOWLEDGEMENTS

This publication was made possible by NPRP grant #8-408-2-172 from the Qatar National Research Fund (a member of Qatar Foundation). The statements made herein are solely the responsibility of the authors.

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